Agenda

* Usecase - [AT&T Churn Prediction]
* Introduction to Logistic Regression
* Step Function
* Sigmoid Function
* Geometric Interpretation of sigmoid function
* logloss function
* sklearn implementation of logistic regression

Assignments

✓ Linear Regression

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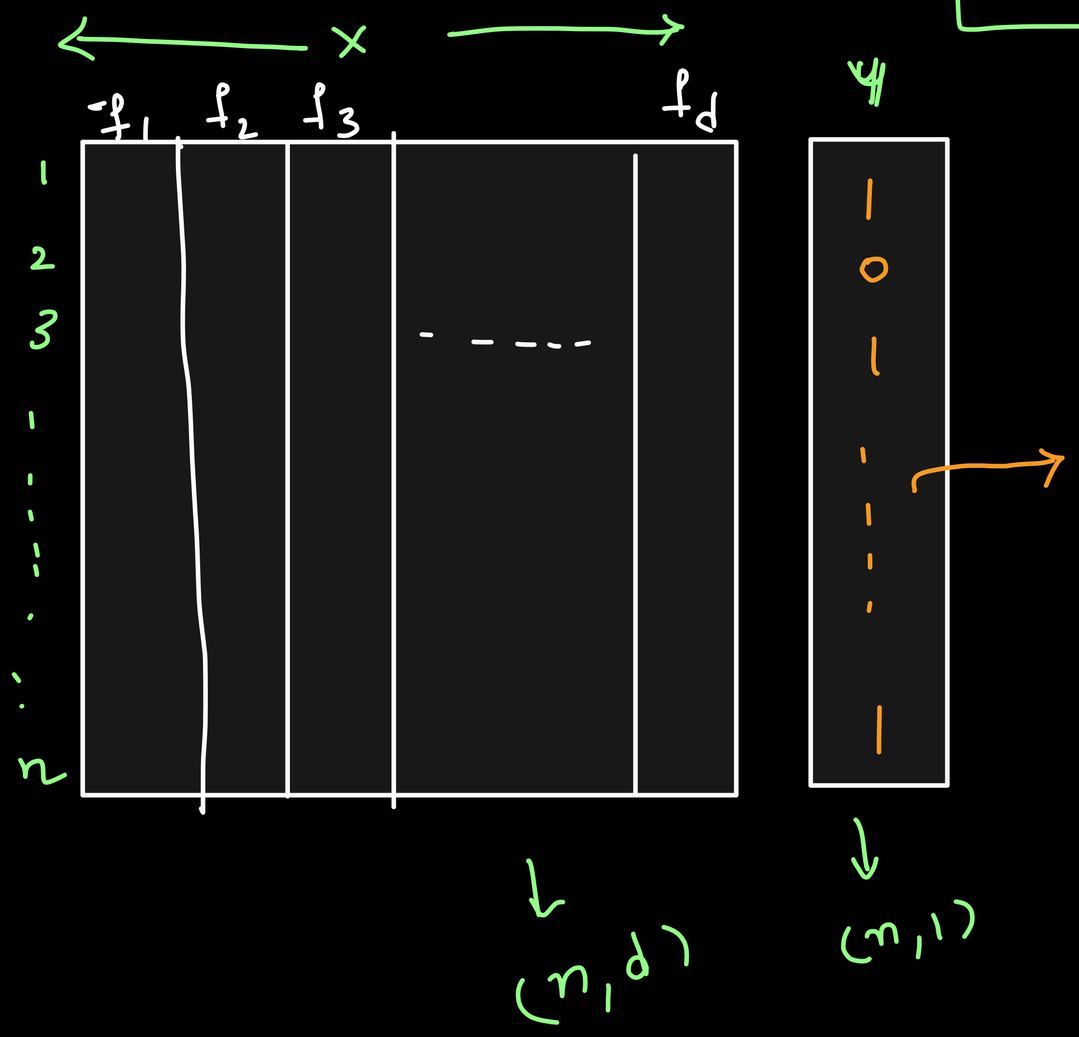
AT & T Churn Prediction

AT & T, one of the leading telecommunications companies of USA, faces the challenge of customer churn, where customers switch to competitors due to issues such as bad network, poor customer service, and so on.

As a Data Scientist, you are tasked to develop a machine learning model to predict whether a customer will be churned or not, based on various parameters, such as state, number of evening calls, number of day calls, etc.

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Intro to Logistic Regression



classification

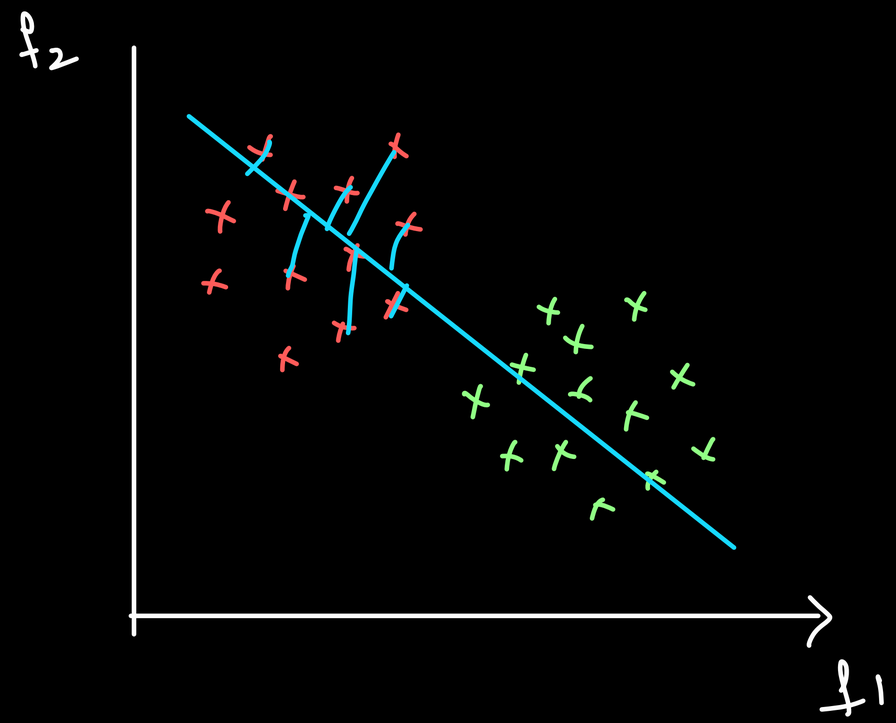
↳ predict discrete o/p "labels" 0/1

categories

→ { Binary classification

0 → Not churned (stay)

1 → churn (leave)

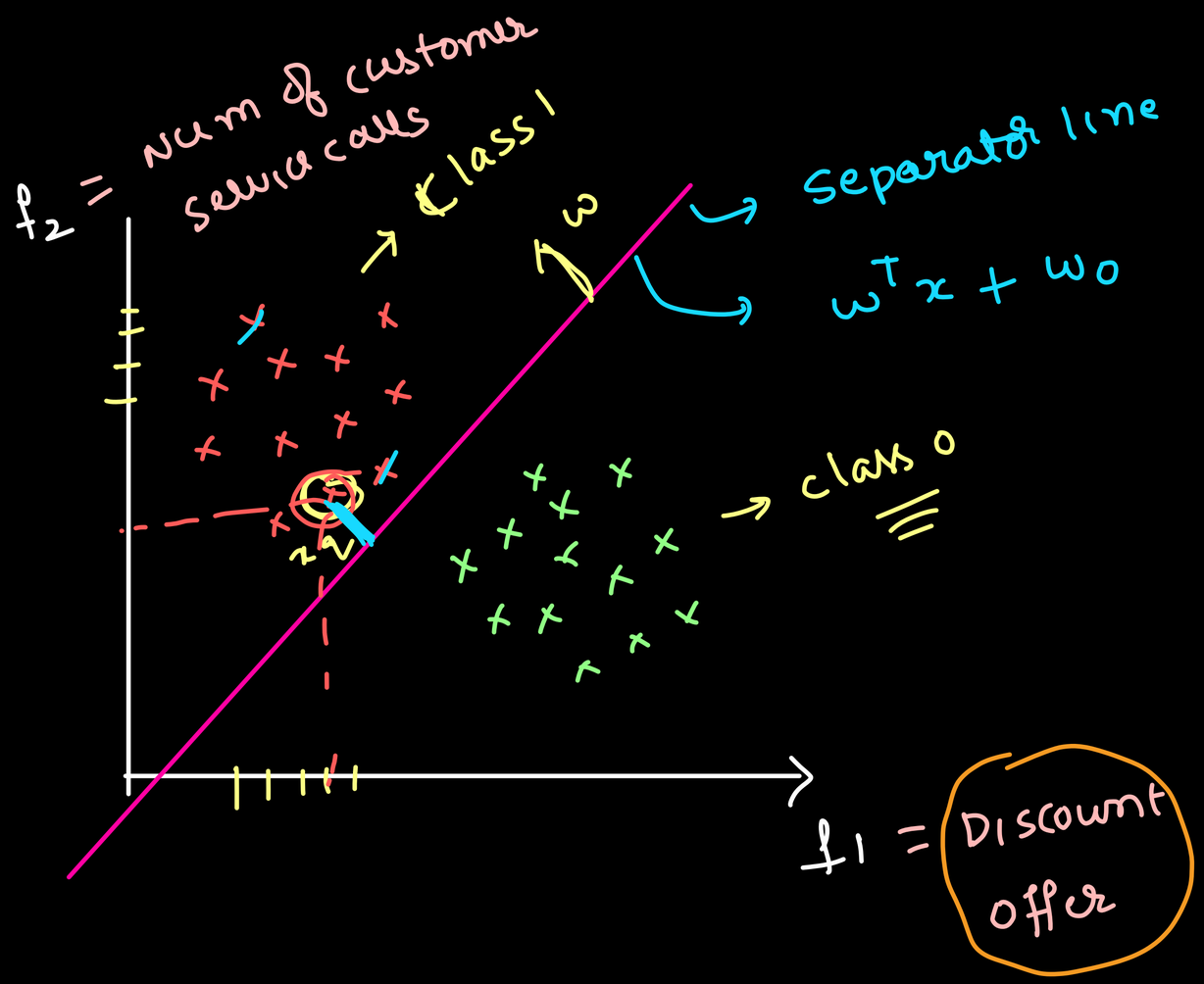


$$\hat{y} = \omega^T x + \omega_0$$

$$\hat{y} \in \mathbb{R}$$

↳ real numbers

line reg → best fit line to minimize MSE



→ log reg finds line / plane / hyperplane

$$w^T x + w_0 = 0$$

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

Eg:

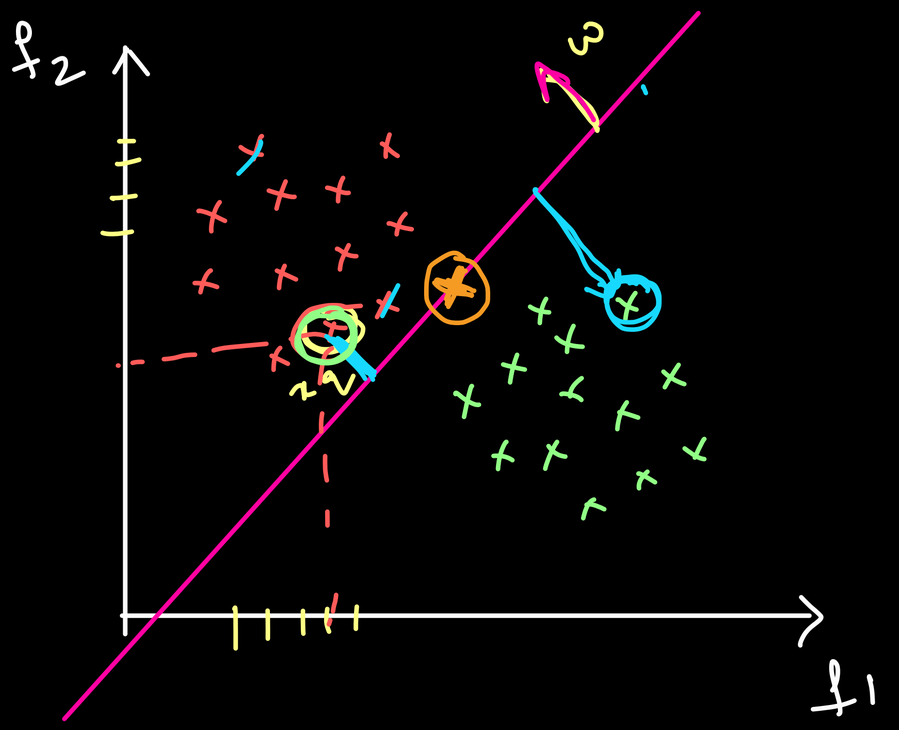
$w_1 = 2$

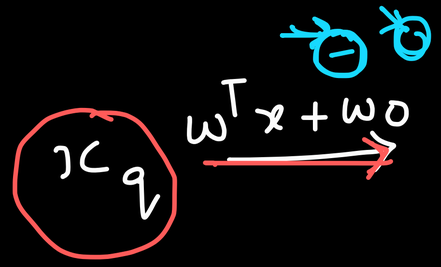
$w_2 = -4$

$w_0 = 1$

→ $2x_1 - 4x_2 + 1 = 0$

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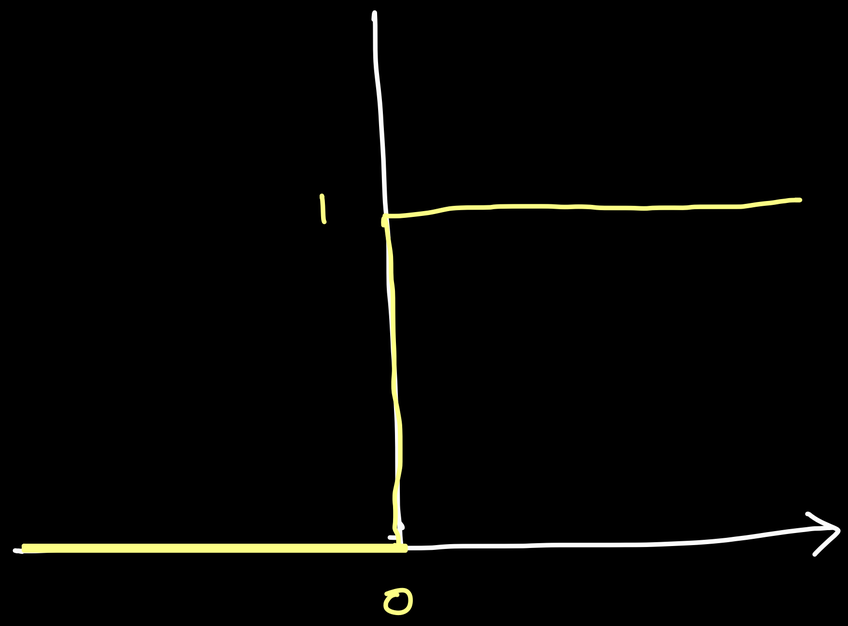




distance from the line

(z)

> 0



$$g(z) = \begin{cases} 1 & \text{distance} \geq 0 \\ 0 & \text{distance} < 0 \end{cases}$$

Step function (Φ)

threshold function

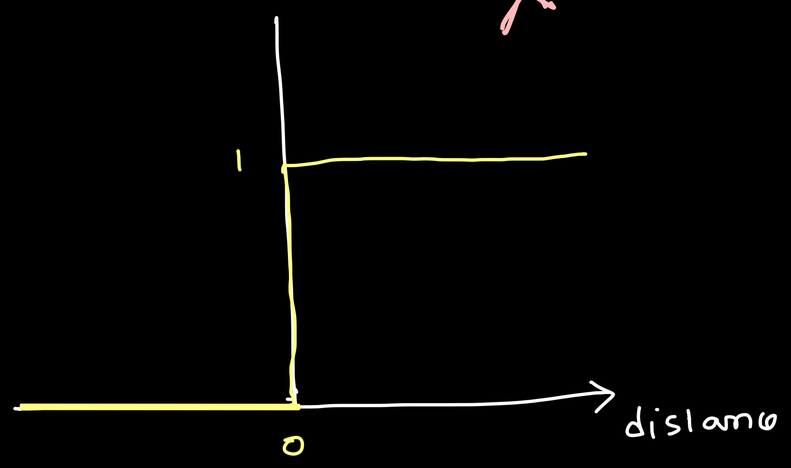
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Process

$x^{(i)}$ $\xrightarrow{w^T x + w_0}$ $z^{(i)}$ (distance from line)

z $\xrightarrow{\text{step function}}$ $\{0, 1\}$

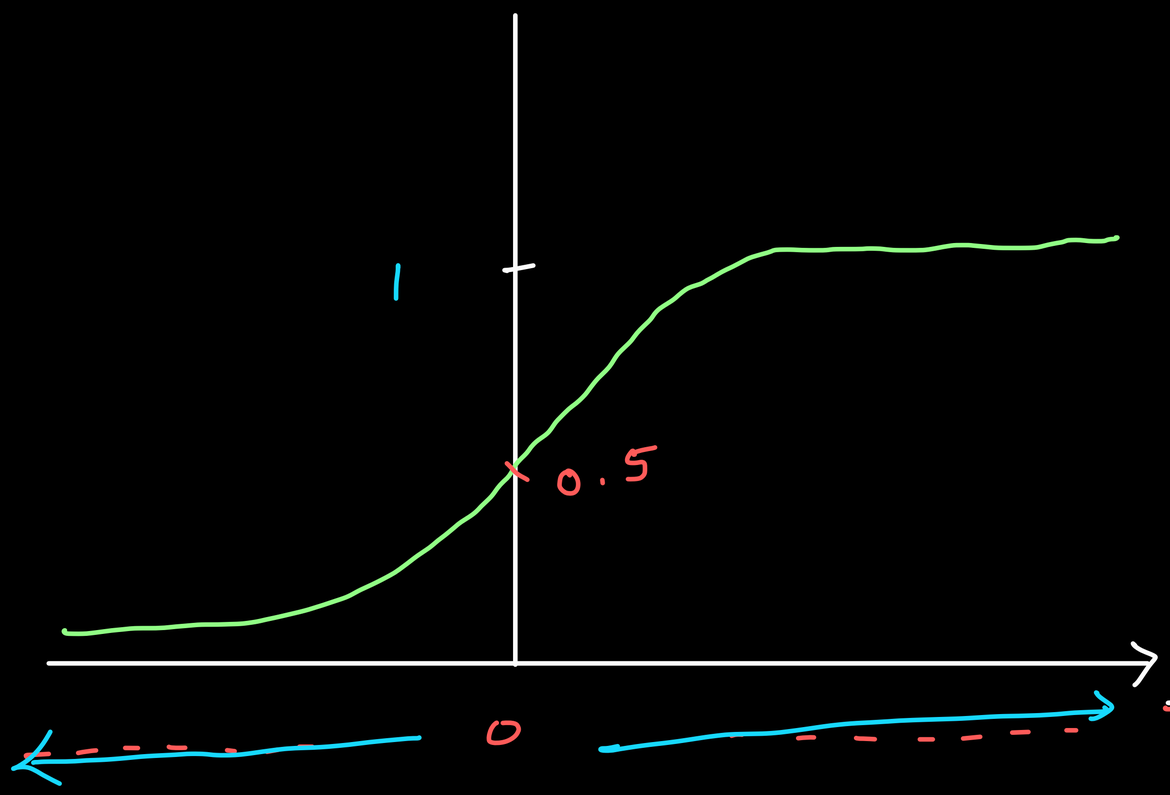
$\times \leq$



→ not differentiable at $z = 0$ ✓

$\frac{\partial}{\partial}$

Sigmoid function (**) ML



$x^{(i)} \xrightarrow{\omega^T x + \omega_0} z^{(i)}$

$z^{(i)} \in (-\infty, \infty) \in \mathbb{R}$

$z^{(i)} \xrightarrow{\text{Sigmoid}} \{0, 1\}$

floor (or ceil)

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$-(-1)$

(1) $z \to \infty$ ↳ very large +ve

$\sigma(z) \to 1 \to 100\%$

(2) $z \to -\infty$ Very large -ve

$\sigma(z) \simeq 0$

$$\frac{1}{1 + e^{0}}$$

(3) $z \to 0$

$= 1/2 = 0.5$

$\sigma(z) = 0.5$

(0 , 1)

Property

customer leaving

$$\sigma(z^{(i)}) = P(y^{(i)} = 1 \mid x^{(i)})$$

Given $x^{(i)}$ sigmoid fn will give you the probability of the data point belongs to class 1

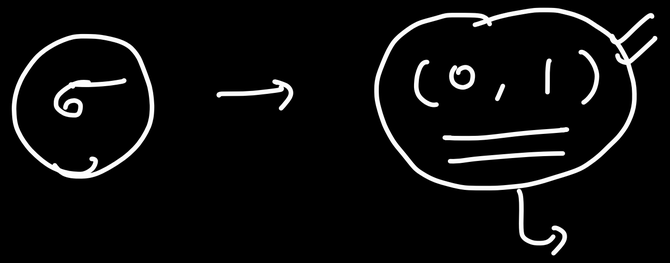
$\sigma(z) \rightarrow 1 \rightarrow 100\%$ prob customer is leaving

$\epsilon(z) \rightarrow 0 \rightarrow 0\%$

$\sigma(z) \rightarrow 0.5 \rightarrow 50\%$ prob customer is leaving

11 → class 1

100% P customer is staying



How can i get either 0 or 1.

$\geq 0.5 \rightarrow 1$ ✓✓

$< 0.5 \rightarrow 0$

$\sigma(z) \geq 0.5 \Rightarrow 1$

$\sigma(z) < 0.5 \Rightarrow 0$

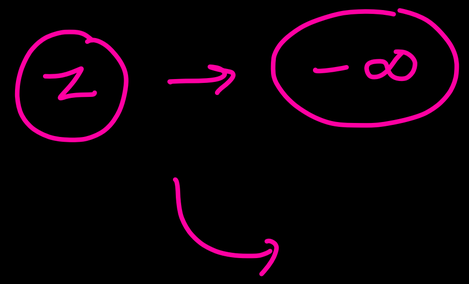
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What happens when the input to the sigmoid function is a very large negative value?

14 users have participated

A	The output becomes negative	0%
B ✓	The output approaches 0	93%
C	The output approaches 1	7%
D	The output becomes undefined.	0%

End Quiz Now

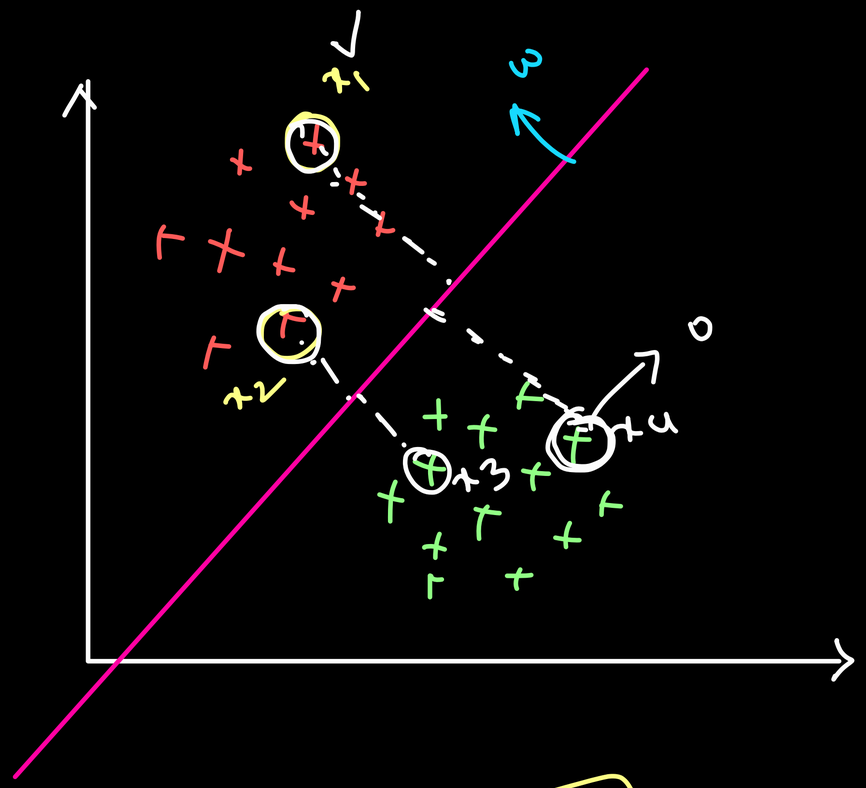


$\sigma(z) \rightarrow \simeq 0$ ✓

10:05 → 10:10 ✓✓

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Geometric interpretation of Sigmoid



Comparision

x_1 → lineeqn

$z_1 = d_1 = w^T x_1 + w_0$

z_1 = 7.4

z_1 → Sigmoid

$\sigma(z_1) \simeq 0.9999$

x_2

$z_2 = d_2 = w^T x_2 + w_0$

$z_2 = 2.2$

$\sigma(z_2) \simeq 0.74$

$w^T x + w_0$

$\|\vec{w}\| \to 1$

$P(y=1 \mid x_1) = 0.99$

99%

very confident

$P(y=1 \mid x_2) = 0.74$

74%

x_3

$z_3 = -1.3$

$\sigma(z_3) = 0.1$

$P(y=1 \mid x_3) = 0.1$

10%

$90\% \rightarrow 0$

x_4

$z_4 = -5.6$

$\sigma(z_4) = 0.02$

$P(y=1 \mid x_4) = 0.02$

98%

$= 1 - 0.02 \Rightarrow 0.98$

$< 0.5 \Rightarrow 0$

$\Rightarrow 2\%$

class 0

Sigmoid

- Smooth & easily differentiable
- Squashes the values b/w **0 to 1** (0, 1)
- Probability of class 1
  - ↳ always be $P(y=1 \mid x^{(i)})$
- Sigmoid function is **monotonically** increasing function

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Logistic Regression

↳ (0 to 1)

continuous

Sigmoid fn → logistic fn

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Loss Function

↳ logloss

$$L(\underline{y}, \underline{\hat{y}}) = \frac{-1}{n} \sum_{i=1}^{n} \left[y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log (1 - \hat{y}^{(i)}) \right]$$

log loss (or)

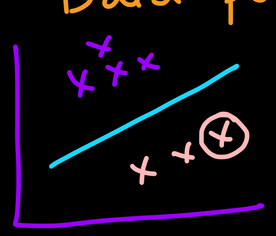
* Binary cross entropy

$y \rightarrow \{0, 1\}$

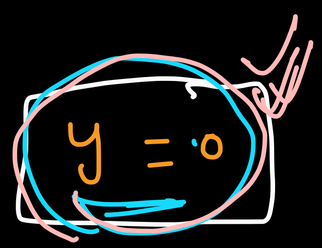
$\hat{y} \rightarrow (0, 1)$ blw 0 to 1

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[\underbrace{y^i \log \hat{y}^i}_{A} + \underbrace{(1-y^i)\log(1-\hat{y}^i)}_{B}\right]$$

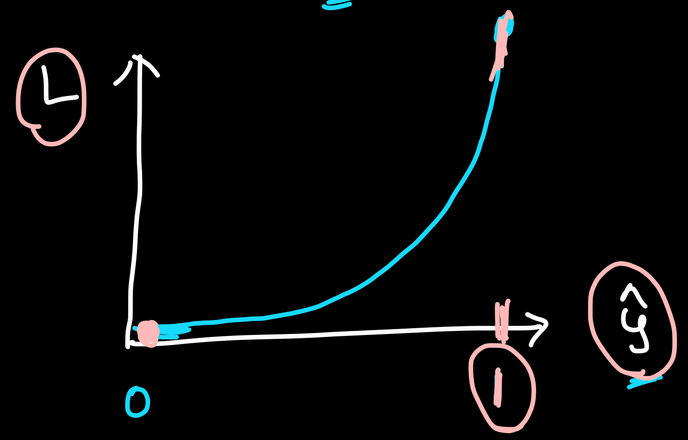
one Data point



Actual

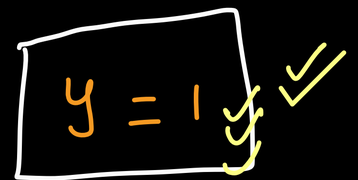


$$L = -\log(1-\hat{y})$$

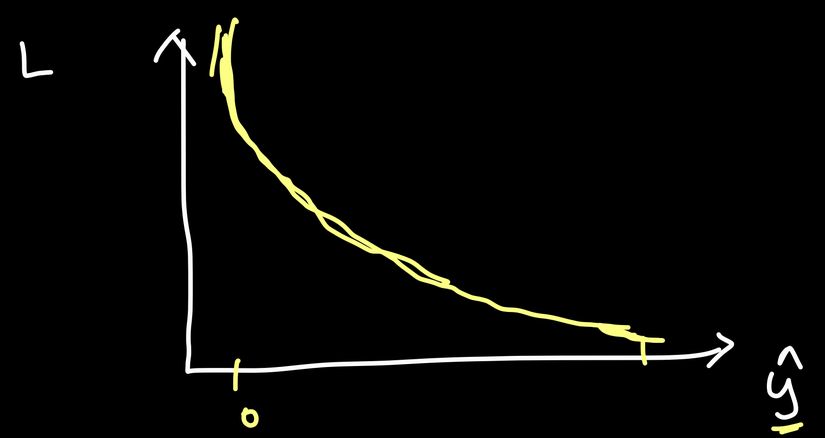


0.999

99.9%



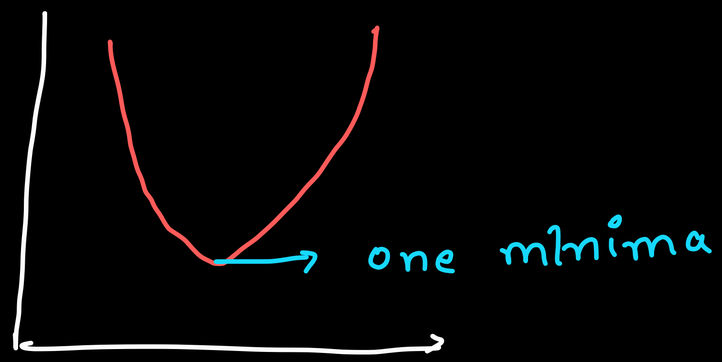
$$L = -\log \hat{y}$$

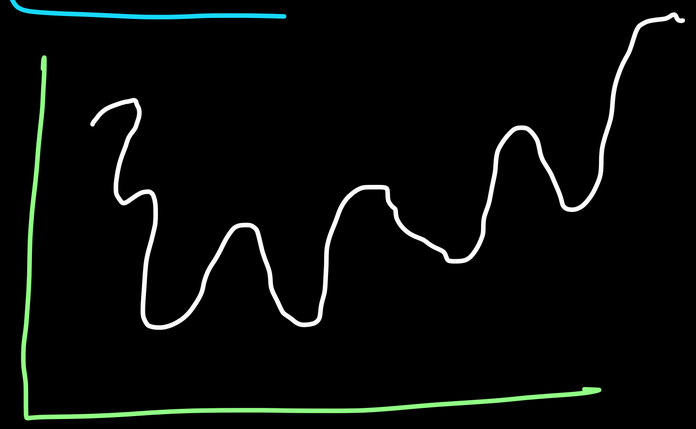


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Q → Why can't we use MSE as the loss for classification?

Convex fn vs Non convex fn



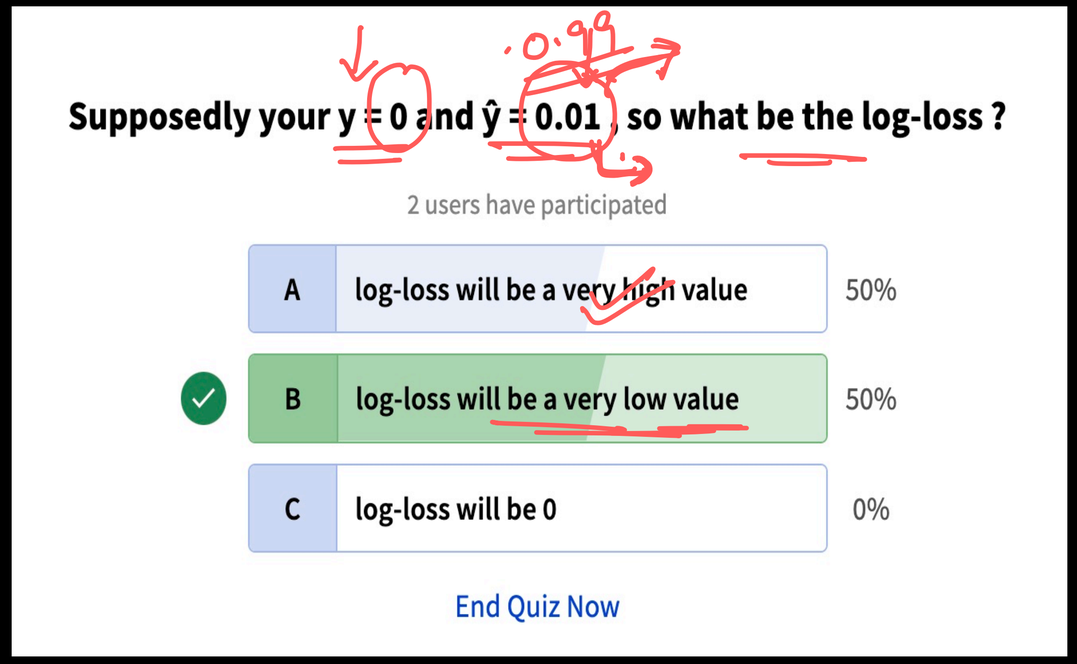


MSE (log. reg) → Non convex

✓MSE (lin reg) → convex

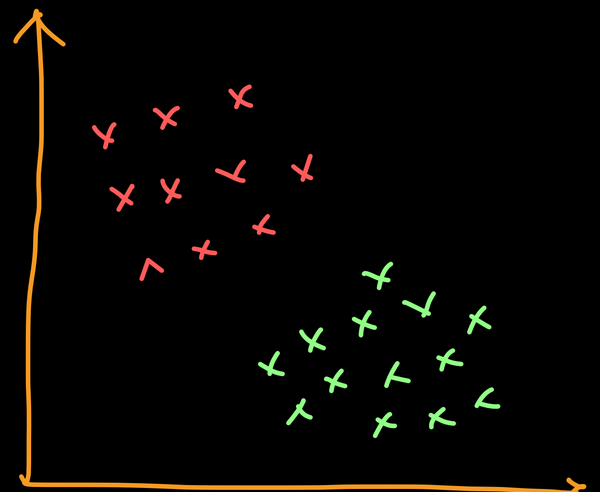
logloss (log. reg) → Convex

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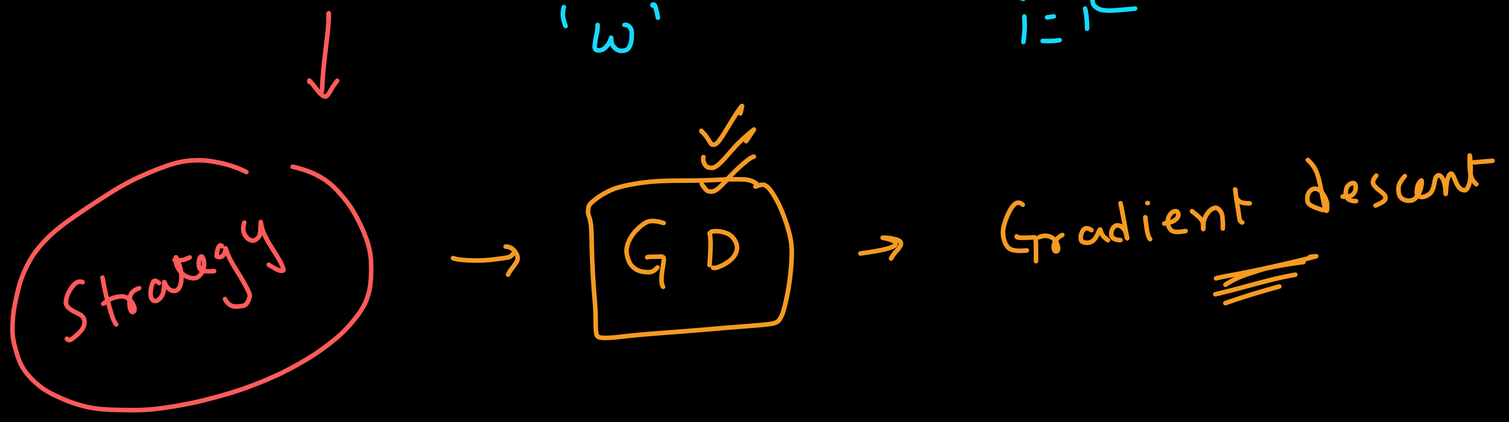

Supposedly your y = 0 and ŷ = 0.01 , so what be the log-loss ?
2 users have participated
A
log-loss will be a very high value
50%
B
log-loss will be a very low value
50%
C
log-loss will be 0
0%
End Quiz Now

Optimization

How do we find the best seperator line ?



$$\underset{w}{\operatorname{argmin}} \; -\frac{1}{n} \sum_{i=1}^{n} \left[y^i \log \hat{y}^i + (1-y^i) \log (1-\hat{y}^i) \right]$$



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Steps

1. Randomly initialize all 'w' weights

2. Repeat until converges {

$$w_j = w_j - \eta * \frac{\partial L}{\partial w_j}$$

gradient

}

Next class

Performance metric

R2-score

Accuracy

y	$\hat{y}$	
1	1	✓
0	1	✗
0	0	✓
1	1	✓
1	1	✓

$$\text{Accuracy} = \frac{\text{correct pred}}{\text{total}}$$

$$= \frac{4}{5}$$

$$= 0.8$$

80%